



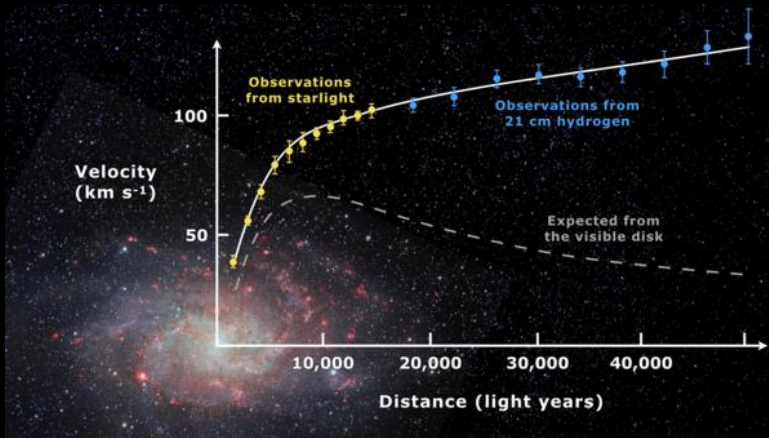
Hunting For Dark Matter with LDMX

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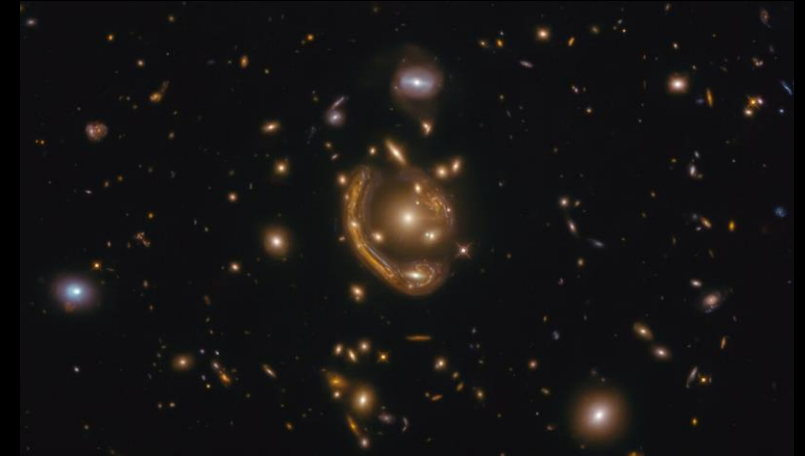
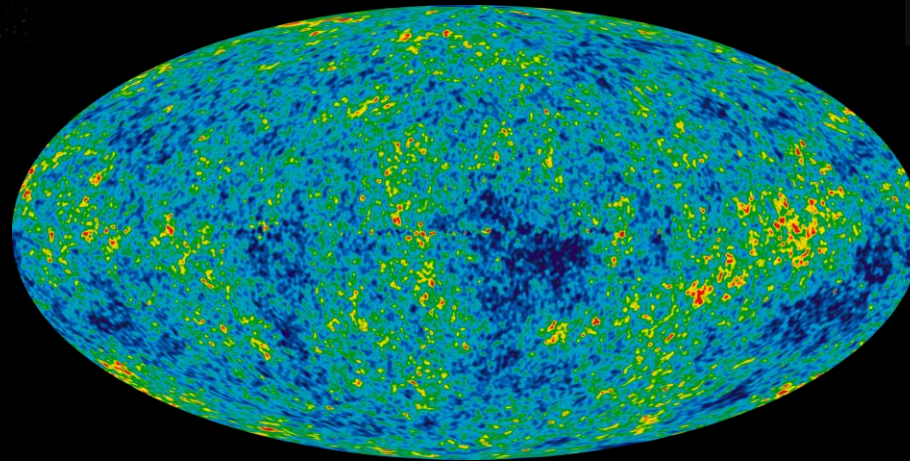
Dark Matter is out there!



Galaxy Rotation Curves

Can look for particle
dark matter here on
Earth!

Cosmic Microwave Background



Gravitational Lensing

Thermal Dark Matter

Mass



- Candidate masses span ~70 orders of magnitude!

Simplifying assumption: Dark matter was in thermal equilibrium in the early universe

- Well motivated
- Dramatically reduced parameter space



- Lots of open parameter space ☺
- The focus of LDMX

- Restricted parameter space ☹

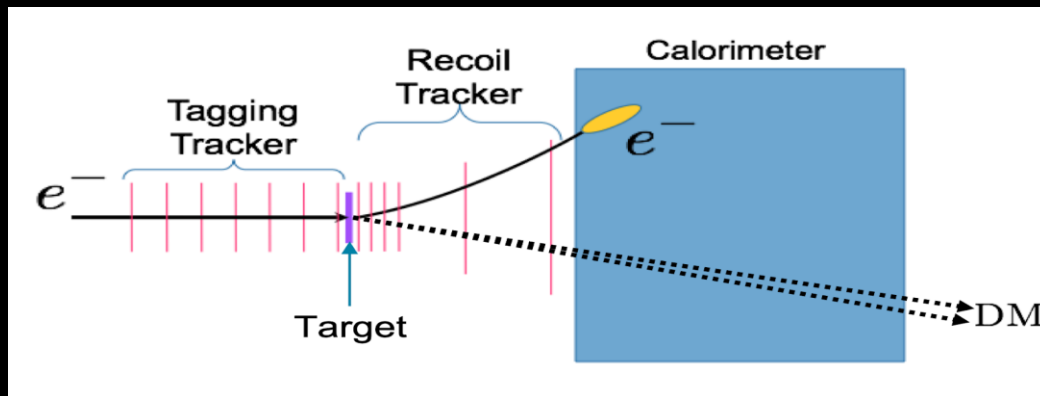
LDMX Concept -> *Missing Momentum*

- Light DM could theoretically interact with SM matter via a “dark photon” (A')
- Potential interaction: Dark Bremsstrahlung
 - Electron slows down around atom and emits an A' instead of a photon!

If $m_{A'} > 2m_{DM}$ then A' decays *invisibly*

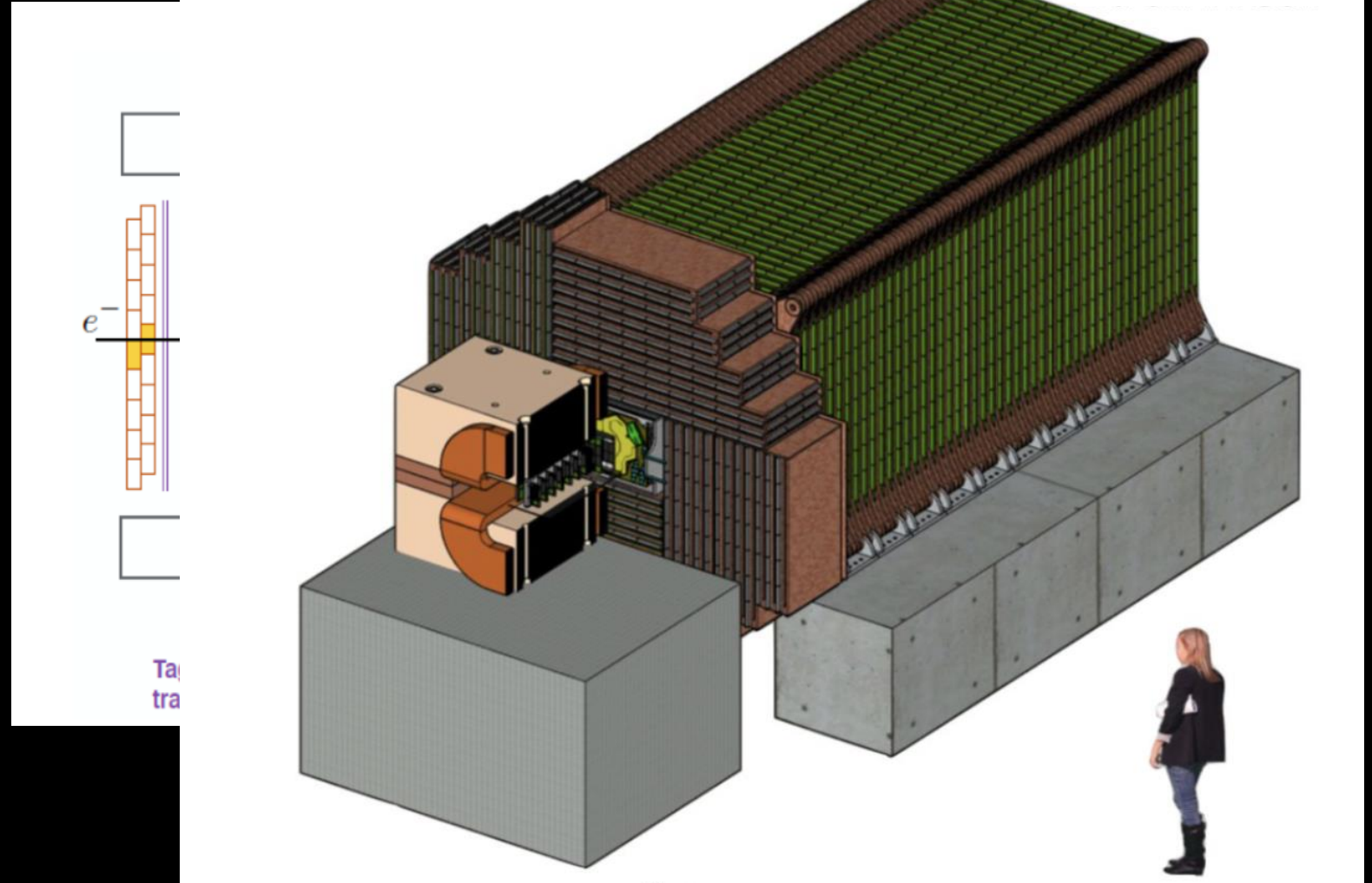
• **Signal Event Signature:**

- single low energy electron with “missing” energy momentum lost to the A'

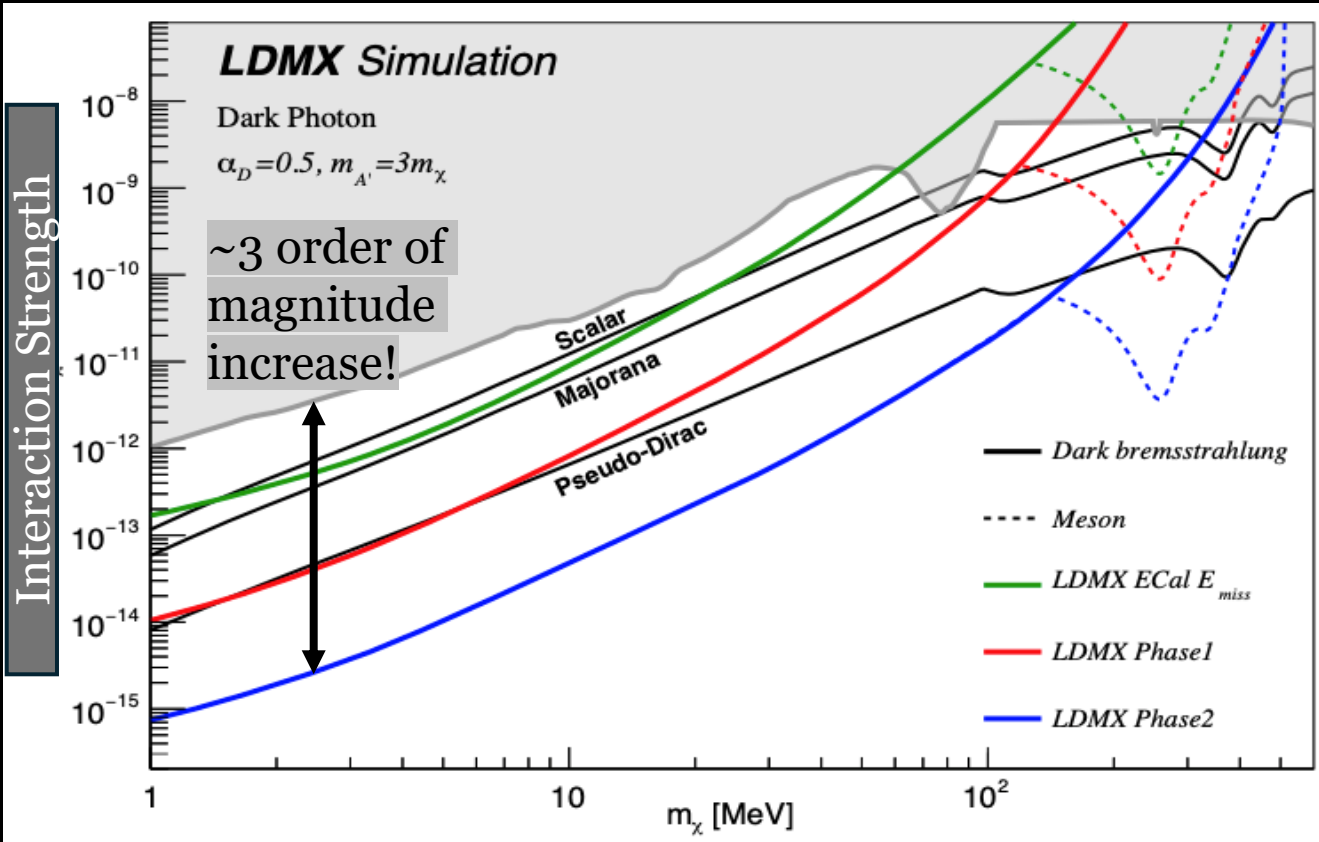


LDMX Design

- High-rate electron beam (from SLAC) at Tungsten target to maximize potential dark-brem events
- Series of detectors and calorimeters to carefully look for missing energy and momentum and veto backgrounds



Missing Momentum Sensitivity



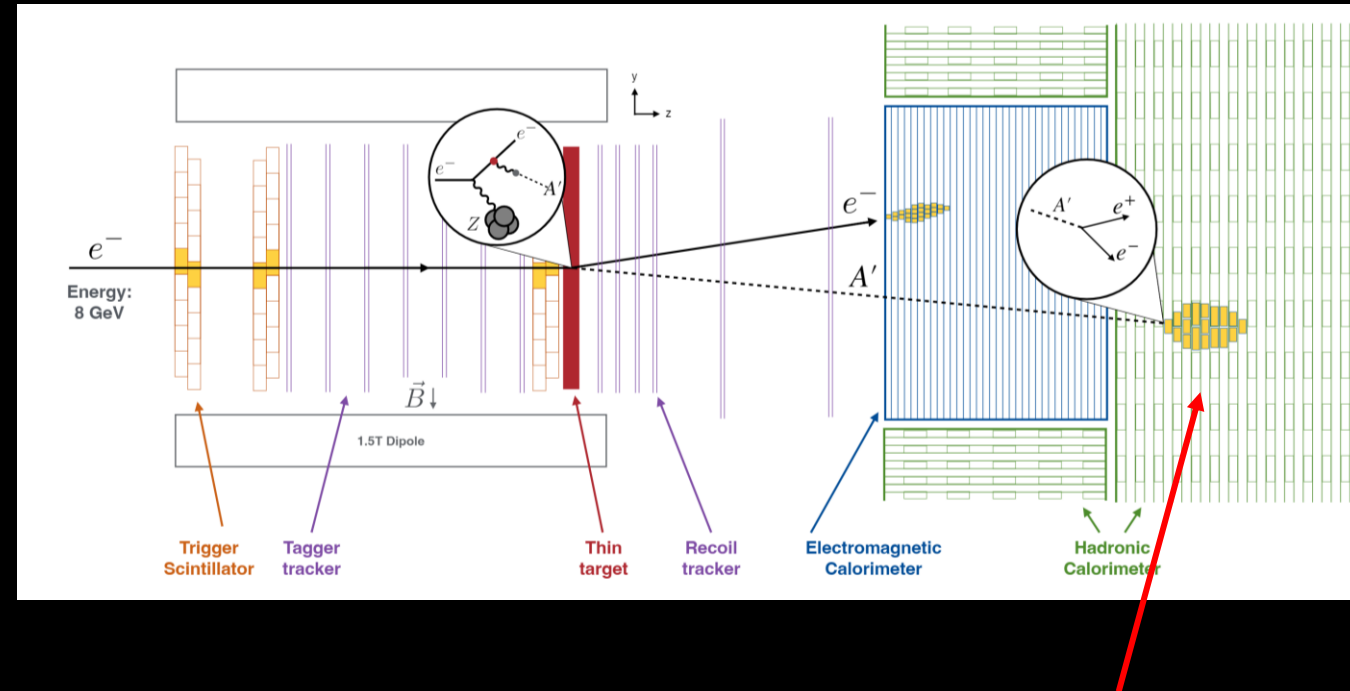
<https://arxiv.org/abs/2203.08192>

And... LDMX has the potential to explore even more physics!

- Within a few weeks of operation, LDMX's missing momentum search will have world-leading results
 - Grey shade is excluded by current experiments
- Due to beam-line delays, most data will be taken at the phase-2 marker in blue.
- Will be huge step forward exploring the Light DM parameter space!

LDMX Visibles -> A Parallel Search

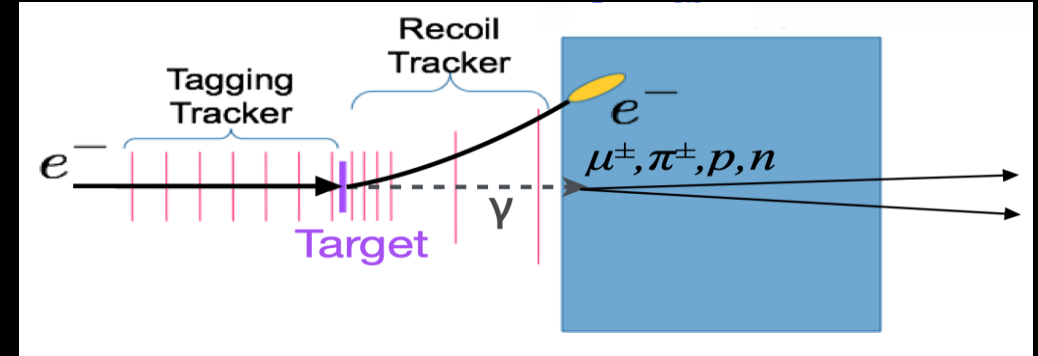
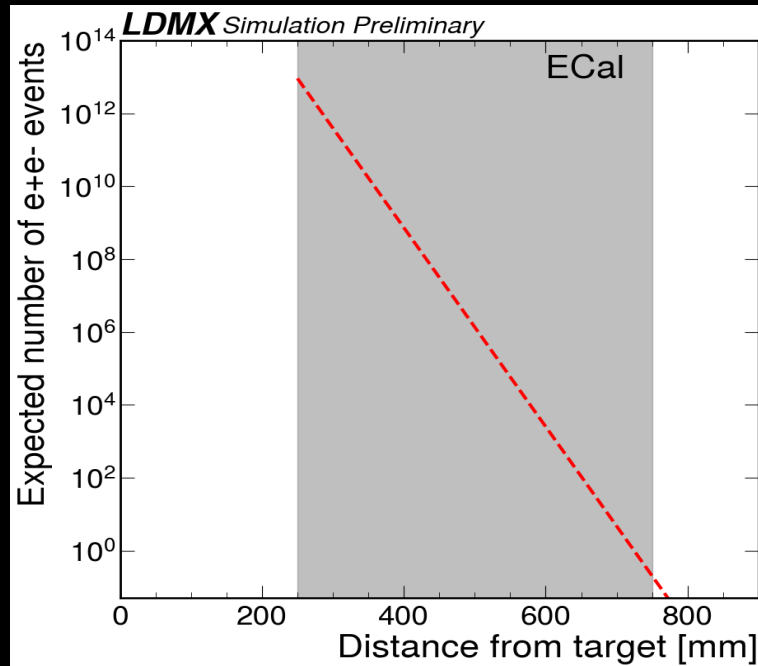
- **What if $m_{A'} < 2m_{DM}$??**
 - A' will decay visibly into SM particles (e^+e^- pair)
 - New signal: appearance of energy deep in Ecal or Hcal with no sign of other SM particles except soft e^-
- More SM processes mimic this signal!
 - My research focus is distinguishing signal vs. backgrounds for visibles!



Displaced E&M Shower

LDMX Visibles -> Backgrounds

- Could in theory look for visible A' in Ecal and Hcal:
 - Simulations revealed irreducible $\gamma \rightarrow e^+e^-$ background
 - Restrict search to Hcal



- Hcal backgrounds:
 - Rare photonuclear events
 - $\gamma \rightarrow \mu^+\mu^-$ events

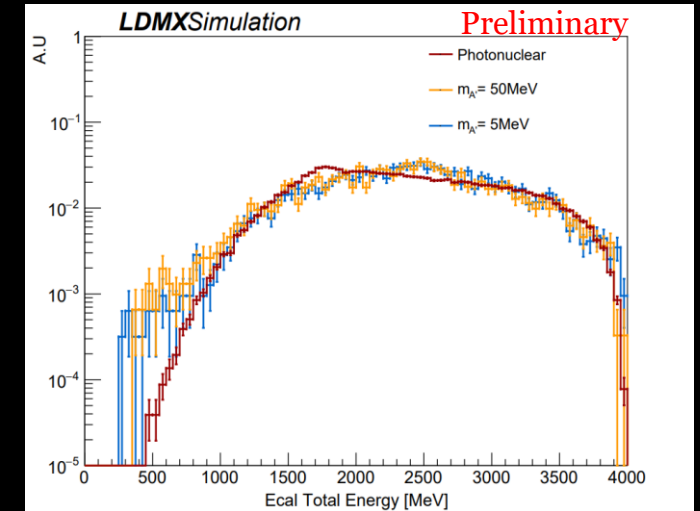
Carefully study
differences in events
features between signal
and background!

Visibles BDT

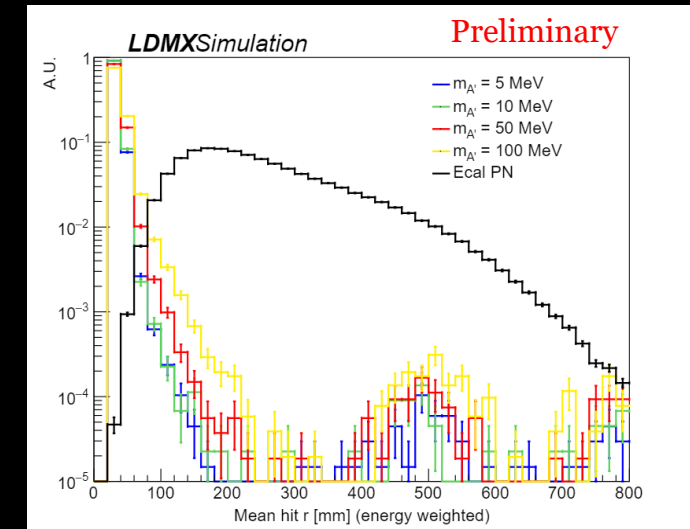
- Some event features have clear difference between signal and background
- In others signal and background very similar
- Could make threshold cuts on single features with good discrimination...

Or even better
combine multiple
high performing
features with BDT!

BDT = Boosted Decision Tree -> Simple but powerful ML model

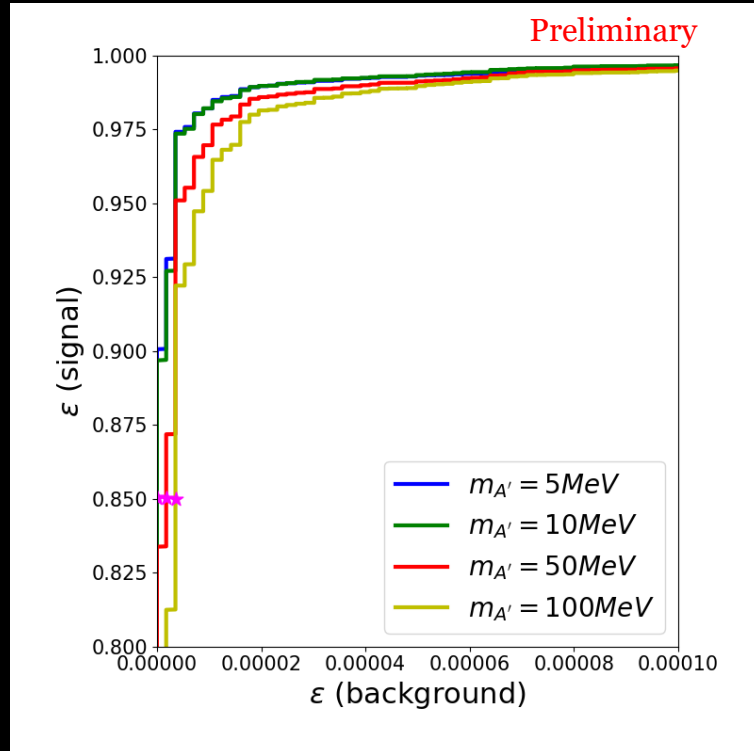


Bad Input Feature ☹️



Good Input Feature 😊

BDT Performance

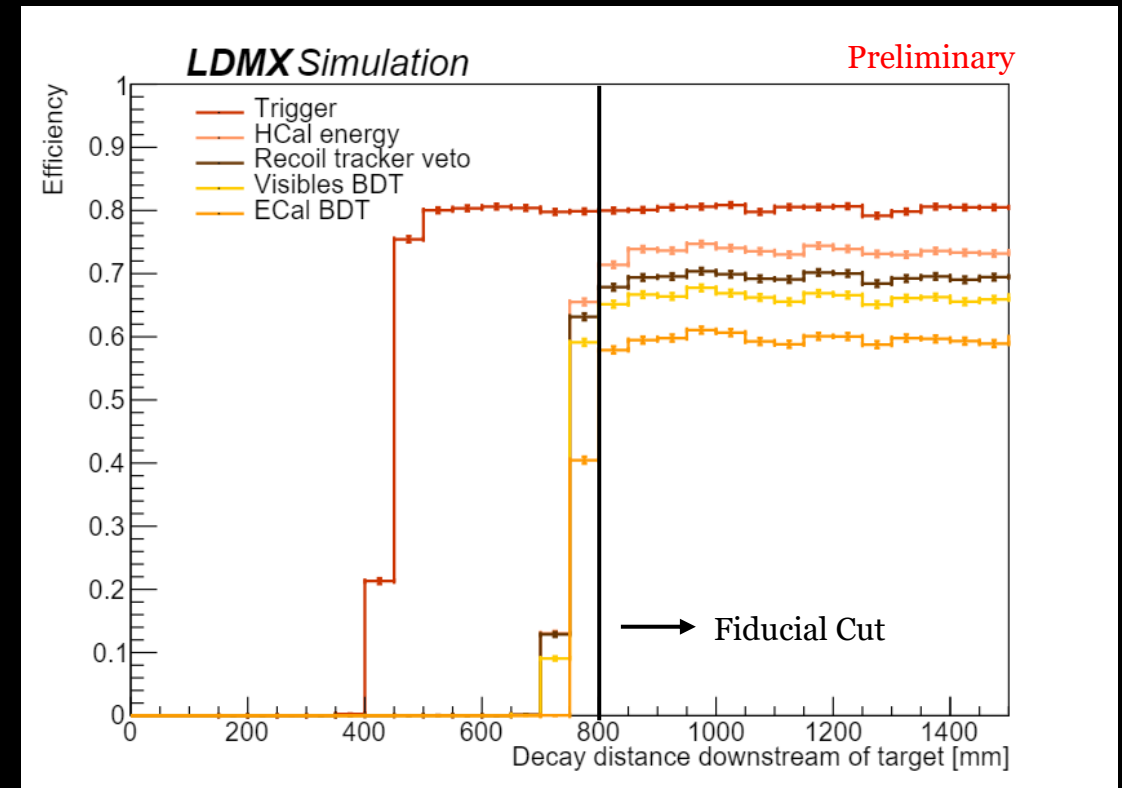


- BDT combines 12 Hcal discriminating features, trained on simulated signal and background events
- Tested against 10^{14} EoT equivalent background sample
 - Preliminary cuts +
Visibles BDT = **189 passing events**
 - Strong signal efficiency

Have some more veto handles to get to zero background!

Further Veto Handles

- Missing Momentum search uses a BDT trained on Ecal features, allows us to incorporate information about recoil e^- shower:
 - Preliminaries + Visibles BDT + Ecal BDT = **5 passing events**
- If we require shower in the Hcal starts beyond first layer, we can reach **zero background events** at 10^{14} EoT. Cool!
 - We maintain high signal efficiency throughout whole cutflow



Conclusion

- LDMX Missing Momentum search will improve Light DM sensitivity by several orders of magnitude
- LDMX Visibles offers complementary search channel with potential for zero background and competitive signal efficiency



Thank You!

Questions?



State-of-the-art LDMX prototype



Our International Collaboration

Image Sources

- https://en.wikipedia.org/wiki/File:Standard_Model_of_Elementary_Particles.svg
- [https://en.wikipedia.org/wiki/Galaxy_rotation_curve#/media/File:Rotation_curve_of_spiral_galaxy_Messier_33_\(Triangulum\).png](https://en.wikipedia.org/wiki/Galaxy_rotation_curve#/media/File:Rotation_curve_of_spiral_galaxy_Messier_33_(Triangulum).png)
- <https://esahubble.org/wordbank/gravitational-lensing/>
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